

Assessment Task 2

[Advanced Data Visualisation]

32146 Data Visualisation and Visual Analytics – Autumn 2025

Minjung Koo
14487805

1. Executive Summary

This report presents a data-driven exploration of the Australian Open's singles finals history, spanning over a century. Using Tableau and Excel, the project visualizes trends across 30 variables including player gender, nationality, seed, set-wise performance, and win rates. The aim is to uncover patterns in player dominance, gender differences, and the impact of seed rankings on championship outcomes.

Key insights include:

- ◆ **Top-seeded players** generally show higher win rates, but several unseeded and low-seeded champions demonstrate that upsets are not uncommon—especially in the women's division.
- ◆ **Country-level dominance** is evident: Australia and the United States lead in total wins, while countries like Belarus and China show strong female representation.
- ◆ **Gender patterns** reveal that while men's champions often hold higher seeds, women's champions are more varied in seed position.
- ◆ **Performance by set** shows that early set wins often predict match outcomes. This trend is visible in parallel coordinate plots and win rate analyses.

The report includes 15+ visualizations—treemaps, geographic maps, scatter plots, word clouds, and parallel coordinates—offering both macro and micro views of performance. The findings provide valuable insights for analysts, coaches, and fans interested in historical tennis patterns and equity in sport.

2. Data Characteristics

Variable	Description	Data Type	Example
Year	Year of the tournament	Discrete	2008
Gender	Gender division	Categorical (Nominal)	Women's
Champion	Name of the champion	Categorical (Nominal)	Serena Williams
Champion Nationality	Champion's nationality	Categorical (Nominal)	USA
Champion Country	Champion's representing country	Categorical (Nominal)	United States
Score	Final match score	Categorical (Nominal)	6–1, 6–2
Champion Seed	Champion's seeding position	Ordinal	1
1st-won	Games won in 1st set	Discrete	6
1st-loss	Games lost in 1st set	Discrete	1
2nd-won	Games won in 2nd set	Discrete	6
2nd-loss	Games lost in 2nd set	Discrete	2
3rd-won	Games won in 3rd set	Discrete	6
3rd-loss	Games lost in 3rd set	Discrete	2
4th-won	Games won in 4th set	Discrete	6
4th-loss	Games lost in 4th set	Discrete	3

5th-won	Games won in 5th set	Discrete	6
5th-loss	Games lost in 5th set	Discrete	4
Runner-up	Name of the runner-up	Categorical (Nominal)	Maria Sharapova
Runner-up Nationality	Runner-up's nationality	Categorical (Nominal)	RUS
Runner-up Country	Runner-up's country	Categorical (Nominal)	Russia
Runner-up Seed	Runner-up's seeding position	Ordinal	5
Total Wins	Number of titles won by player	Discrete	6
Total Games Played	Total number of matches played	Discrete	9
Win Rate	Match win rate	Continuous	0.88
1st Set Win Rate	Win rate in the 1st set	Continuous	1.0
2nd Set Win Rate	Win rate in the 2nd set	Continuous	0.8
3rd Set Win Rate	Win rate in the 3rd set	Continuous	1.0
4th Set Win Rate	Win rate in the 4th set	Continuous	0.6
5th Set Win Rate	Win rate in the 5th set	Continuous	0.5
Total Win Rate	Overall win rate across all sets	Continuous	0.88

Data Source and Scope

- ◆ Dataset covers Australian Open singles finals from **1905 to 2025 in men's and women's tournaments.**
- ◆ Sourced from compiled historical match data (structured manually or collected via online archives, using Total of **30 variables** in the analysis)

Missing Data Handling

- ◆ **Champion Seed**
Unseeded players or Missing value → **encoded as 0** to represent unseeded status
 - Clearly documented and retained for analysis in scatter plots and seed-related trends
- ◆ **Set-by-Set Win Rate**
Missing Value → **left blank to preserve accuracy** in visualizations like parallel coordinates
- ◆ **Match Duration (mins)**
Only 4 valid entries existed → **removed** from dataset to maintain representative variable distribution

Notable Transformations

The dataset was significantly enhanced by manually calculating and adding several key performance fields to enable more granular analysis:

- ◆ 🏆 **Total Wins**
 - Number of matches won by each player across all years (Discrete)
- ◆ 🎾 **Total Games Played**
 - Total number of matches played per player, used for win rate calculations (Discrete)
- ◆ 📊 **Win Rate**
 - Calculated as (Total Wins / Total Games Played); gives an overall match-level win ratio (Continuous)
- ◆ 🧩 **Set-by-Set Win Rates**
 - Added 1st Set Win Rate, 2nd Set Win Rate, ... up to 5th Set Win Rate, based on set-level game outcomes (Continuous)
 - Useful for detecting consistency or drop-off patterns across sets
- ◆ 📈 **Total Win Rate**
 - Aggregated average of all set-wise win rates per player (Continuous)

These fields were not originally present in the dataset and were created manually through data wrangling and formula-based transformations.

3. Introduction

The Australian Open is one of the four Grand Slam tennis tournaments, with over 120 years of championship history. This report explores performance patterns among singles champions and runner-ups, focusing on gender, seed rankings, country representation, and match outcomes.

The main objective of this project is to:

- ◆ Identify **trends in player performance** based on seeding and gender
- ◆ Compare **national dominance** in men's and women's tournaments
- ◆ Examine **win rates and set-wise performance** to uncover key competitive patterns

To achieve this, we utilized Tableau and Excel to create over 25 visualizations, including tree maps, scatter plots, choropleth maps, and parallel coordinate plots. These visualizations help highlight both overall trends and deeper insights related to individual players and tournament outcomes.

By investigating the relationship between **seed rankings and win rates**, and by comparing outcomes across gender and country, this analysis offers a clearer picture of what contributes to success in elite tennis. The results may provide valuable context for analysts, athletes, sports federations, and enthusiasts alike.

4. Visualisation & Analysis

4.1 Global Overview of Participation

- Visualisation Types: Choropleth Maps, Map Charts

► Participated Countries

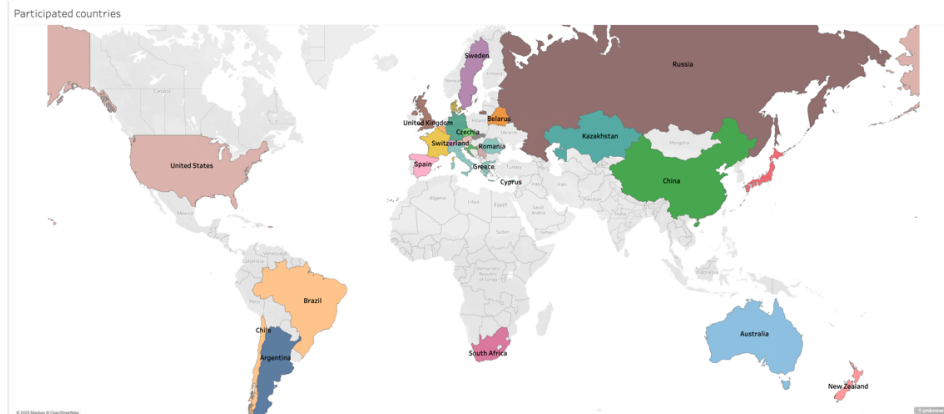


Figure 1: Overview of participated Countries in the world

⇒ This choropleth map visualizes the geographic spread of countries that have produced at least one Australian Open champion or runner-up, highlighting the global appeal and inclusivity of the tournament.

4.2 Champion & Runner-up Distribution

- Visualisation Types: Choropleth Maps, Map Charts

► Champion vs Runner-up (Map view)

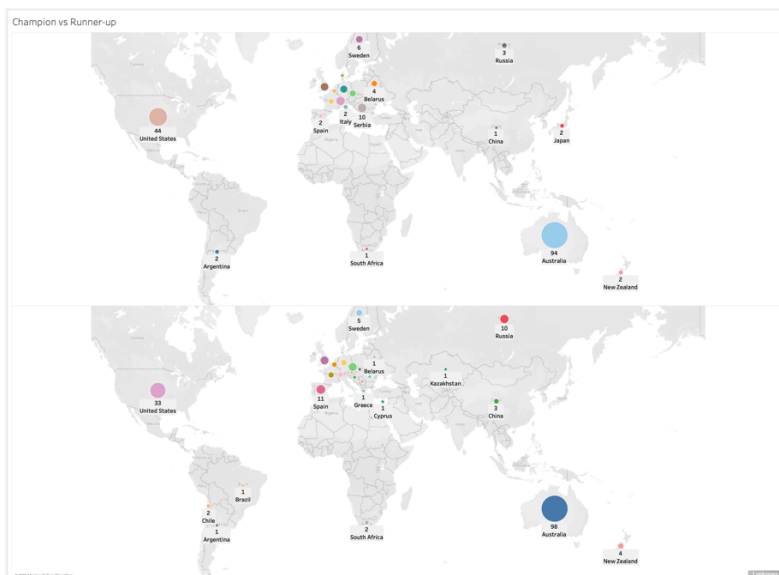


Figure 2: Champion vs Runner-up (Map Overview)

⇒ The top map shows the number of champions by country, while the bottom displays runner-ups. Circle size denotes total counts. Australia dominates in both, followed by the U.S., Switzerland, and Serbia.

4.3 Gender-based Analysis

► Men's and Women's Championship Timeline (by Year)

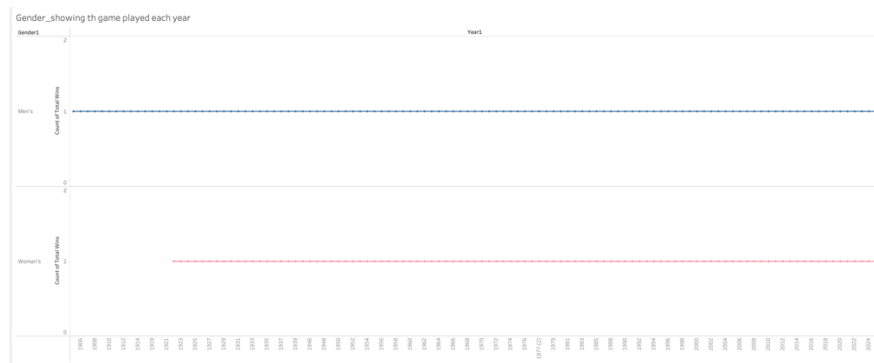


Figure 3: Historical Timeline of Men's and Women's Championships

⇒ The Australian Open men's singles championships have been held consistently since 1905, whereas the women's singles match began in 1922. This highlights a **17-year historical gap** in official gender participation at the tournament.

► Match History Overview (Yearly Trends)

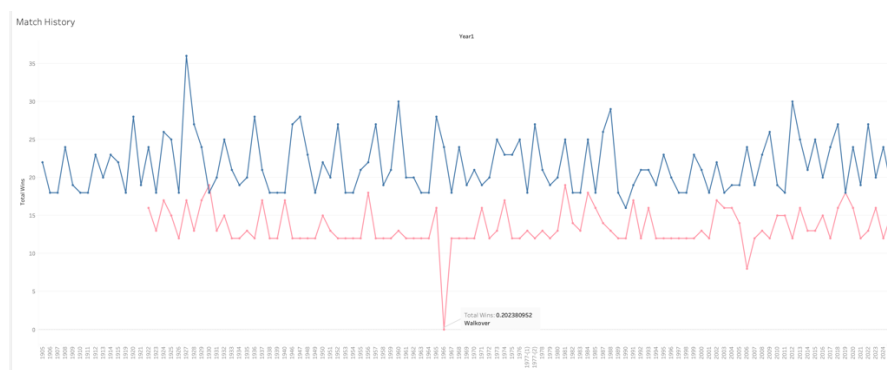


Figure 4: Yearly Match History Trends (Including Notable Events)

⇒ This graph shows the number of matches played each year for both men's and women's singles. Notably, in **1966**, Margaret Court won the championship by **walkover**, meaning her opponent withdrew—this is reflected as a dip to nearly zero in the data.

► Champion vs Runner-up by Gender

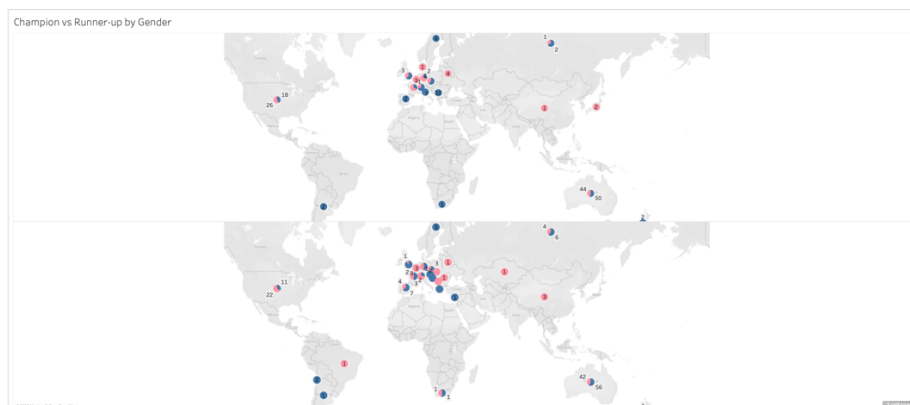


Figure 5: Global Distribution of Champions (top) and Runner-ups (bottom) by Gender

⇒ This map visualisation displays the number of champions and runner-ups by gender across different countries. The **top map** highlights **champion distributions**, while the **bottom map** shows **runner-up distributions**. **Blue** represents **men**, and **pink** represents **women**, enabling a quick visual comparison of gender participation by country. **Each dot includes an embedded pie chart** that visualises the **gender ratio** of champions or runner-ups in that specific country, allowing for a clearer understanding of dominance or balance between male and female players within each nation.

► Total Wins by Gender (Women vs Men)



Figure 6: Total Championship Wins by Gender & Country

⇒ These maps illustrate the total number of championship titles by gender across countries. Australia and the U.S. are standout leaders in both men's and women's categories, with a strong balance. Notably, Belarus only appears in the women's map.

💡 Insights Summary

- ◆ **Australia and the U.S.** are dominant in **both men's and women's categories**, showing balanced excellence.
- ◆ **Belarus** appears **only in the women's map**, suggesting a **gender-specific strength**.
- ◆ The visual highlights **global disparities** in gender contributions to championship titles.

► Total Runner-up by Gender (Women vs Men)



Figure 7: Runner-up Distribution by Gender & Country

⇒ Runner-up maps reveal similar trends, but with broader gender distribution across more countries. Countries like Spain and Russia show higher runner-up appearances than championship wins, especially among women.

► Gender Dominance by Country (Proportional)

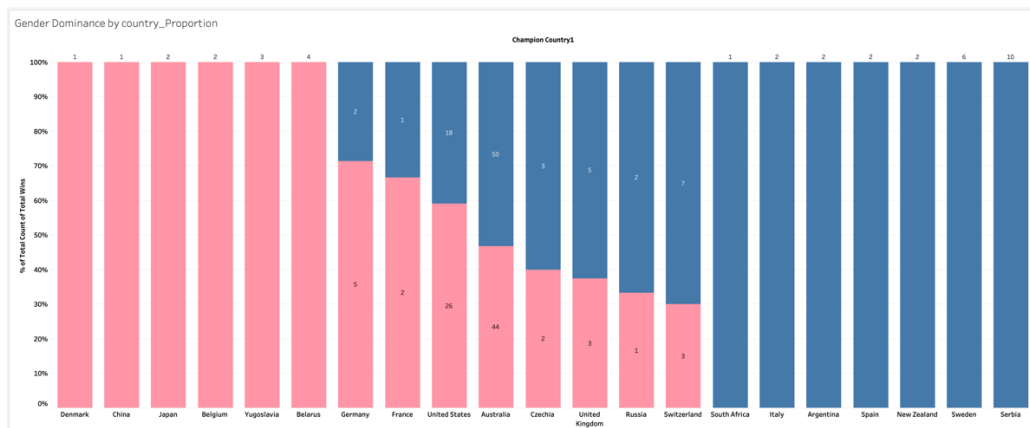


Figure 8: Gender-Based Championship Dominance (Proportional View)

⇒ This 100% stacked bar chart shows the proportion of male and female champions by country, revealing gender-specific strengths across national tennis histories with the number of championships won.

► Gender Dominance by Champion and Runner-up Country (*Map view*)

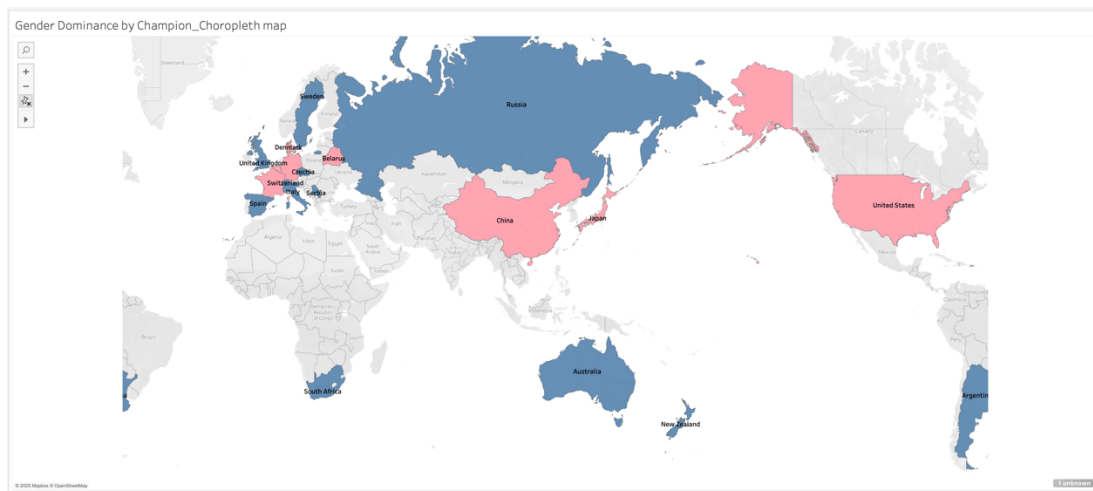


Figure 9: Champion Gender Dominance by Champion Country

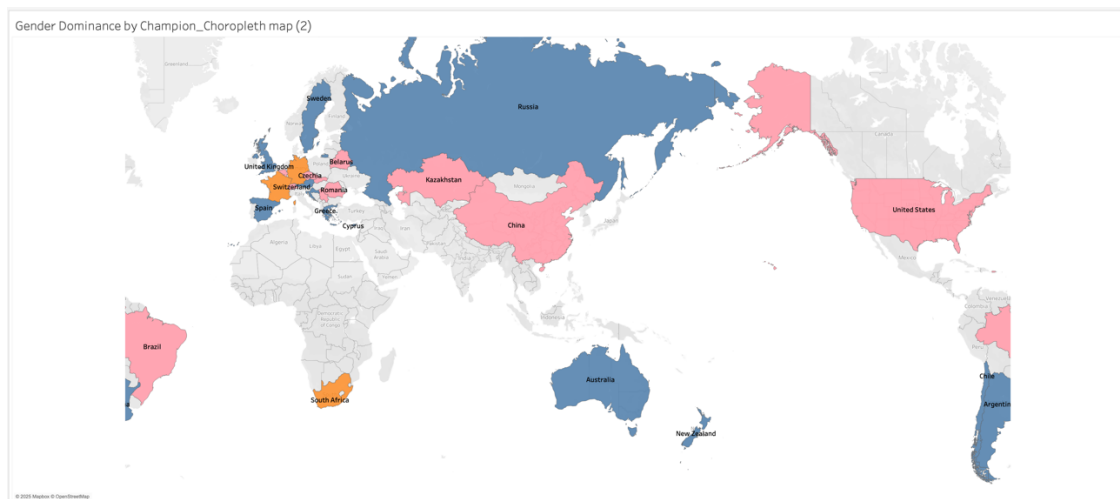


Figure 10: Runner-up Gender Dominance by Runner-up Country

⇒ **Blue** represents countries with more **male winners**, and **pink** indicates **female dominance**. Notably, **orange** highlights countries with **equal numbers of male and female winners or runner-ups**, such as **Germany, South Africa, and France**, as shown in **Figure 12**. The extended view in the runner-up map provides deeper insight into **balanced gender achievements** across the tournament history.

💡 Insights Summary

- ◆ **Male dominance** is evident in countries like **Russia, Serbia, and Switzerland**, while **female dominance** appears in the **U.S. and China**.
- ◆ **Balanced gender outcomes** are found in **Germany, France, and South Africa**, suggesting stronger gender parity.
- ◆ **Runner-up map (Figure 12)** shows broader global participation and **more diverse gender distribution**, especially beyond top-winning countries.
- ◆ Countries with **high female runner-up counts** may have potential to produce future champions.

4.4 Top Player Analysis

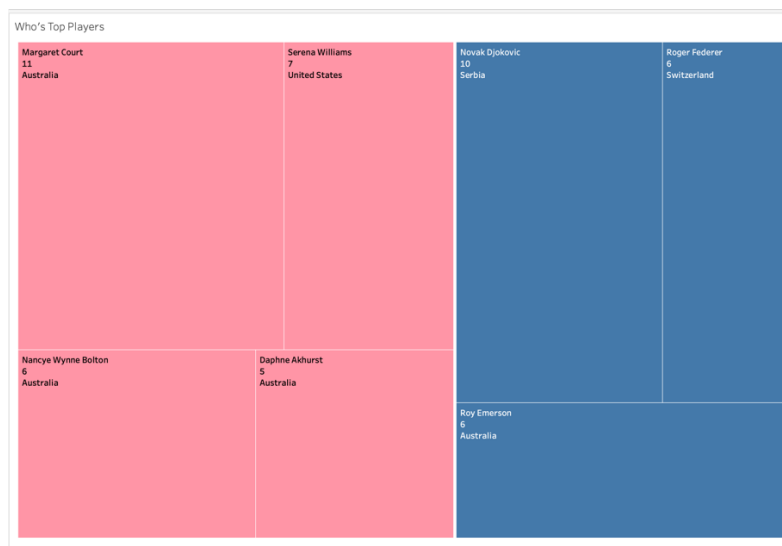


Figure 11: Top players with 5 or More wins, Visualised by Gender and Country

⇒ The treemap highlights players with **five or more championship wins**, sized by their total titles. Notable figures include **Margaret Court** (11 titles, Australia), **Novak Djokovic** (10 titles, Serbia), and **Serena Williams** (7 titles, USA). The chart visually contrasts male and female dominance among top achievers, showing the number of championships won and players' countries, with **Australia** being home to several legendary champions.

► Win Rate Patterns across Sets for Top Players

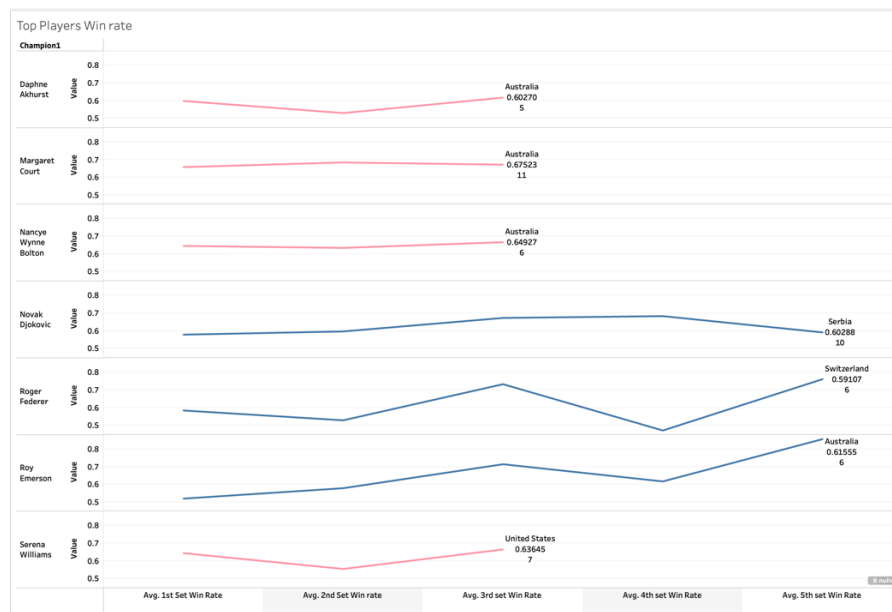


Figure 12: Set-by-Set Win Rate Patterns of Top Players

⇒ This line chart above illustrates the set-by-set win rate patterns of top players who have secured five or more championships. While all are high achievers in terms of titles, the graph reveals that their **overall**

win rates are not exceptionally high, often hovering between 0.60 and 0.68. Notably, **female players' data ends at the third set**, as women's matches are best-of-three, while **male players extend to the fifth set** (best-of-five), reflected by the additional two points. This distinction highlights the **structural difference in match formats between genders**, impacting both the length and dynamics of performance trends.

► When Top Players Played (Gantt Chart)

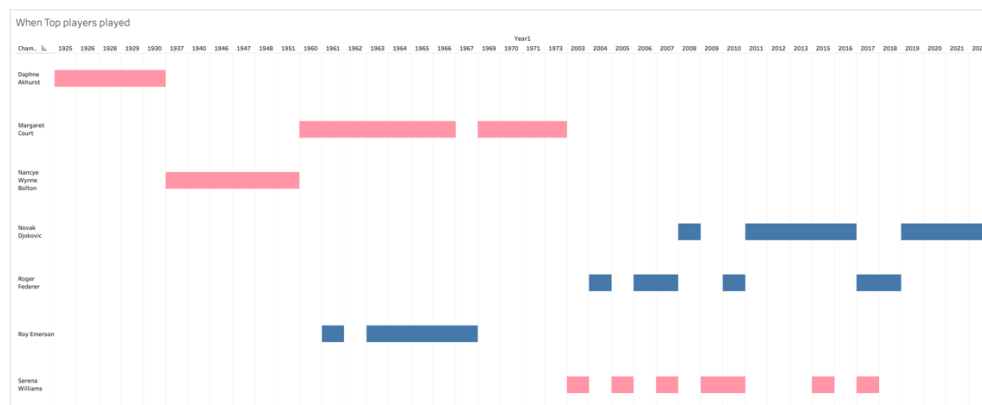


Figure 13: Championship Winning Years of Top Players (by Gender)

⇒ This Gantt chart illustrates the active championship years of top players, colour-coded by gender (pink for Women and blue for Men). It highlights temporal differences between male and female champions, with female players appearing more frequently in earlier decades and male players dominating recent years.

4.5 Win Rate Analysis

► Total Win Rate vs Seed (with Gender-wise Trendline)

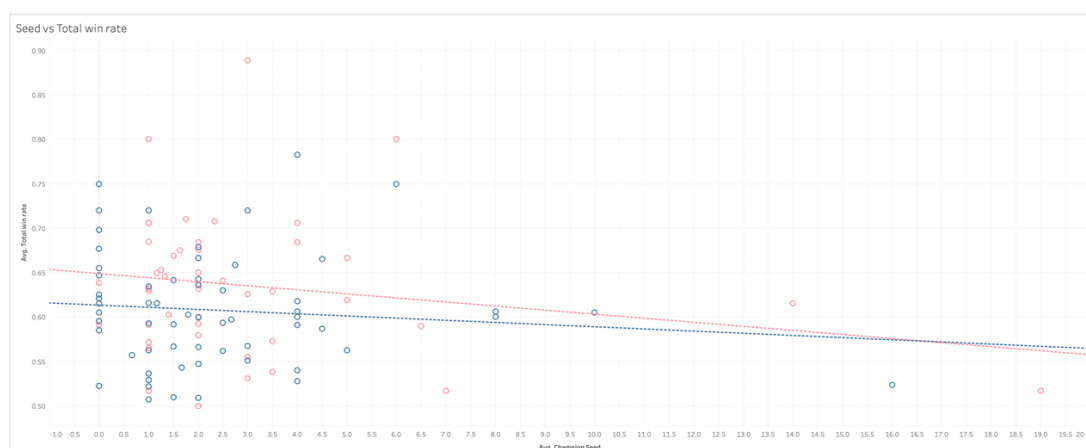


Figure 14: Champion vs Runner-up Dominance by Country

⇒ This scatter plot investigates the relationship between **champion seed rankings** and their **average total win rates**, categorized by gender.

- ◆ **X-axis:** Average Seed (lower = higher ranked)
- ◆ **Y-axis:** Average Total Match Win Rate
- ◆ **Each dot is a player**, coloured by gender (pink for women and blue for men)

Both trendlines show a **slight negative slope**, indicating that players with **higher seeds (i.e., lower numerical values)** tend to have better win rates. This aligns with expectations that top-seeded players generally perform better.

🔍 However, the relationship is not strongly linear — the data includes several **outliers**, such as:

- ◆ **Low-seeded players with unusually low win rates**, possibly due to early exits despite seeding.
- ◆ **High-seeded players with unexpectedly high win rates**, suggesting **overperformance** or breakout performances beyond what their seed predicted.

💡 These outliers emphasize that **seeding alone doesn't guarantee performance**, and other factors (like mental resilience or match conditions) may play a role in championship outcomes.

◆ Note on Seed Value '0':

Seed values labelled as '**0**' represent either **unseeded players** or instances where **seeding data was missing**. These are retained for completeness but should be interpreted with caution, as they don't reflect official seed rankings.

► Country-wise Win Rate by Gender

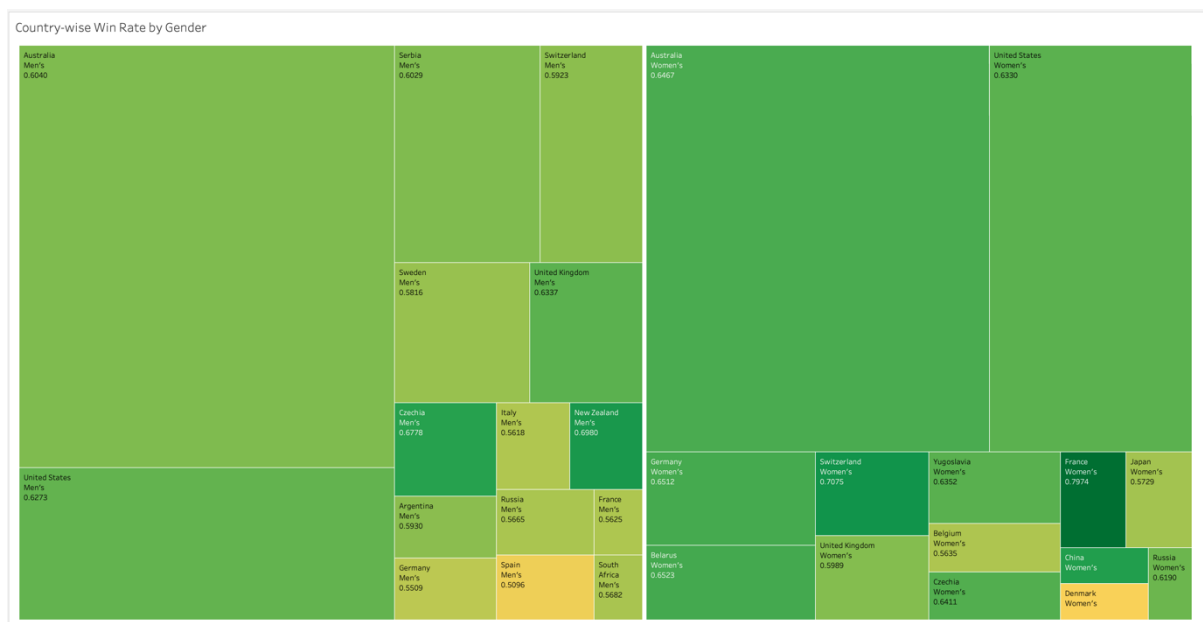


Figure 15: Country-wise Average Win Rates by Gender (Tree Map)

⇒ This tree map compares **average win rates** of champions by country and gender. Each box size reflects the **volume of matches or wins**, while colour intensity indicates **performance (win rate)**.

- ◆ 🇦🇺 Australia dominates both genders, showing **high win rates** and large match volumes.
- ◆ 🇺🇸 The United States shows **strong female performance**, with women's win rate slightly higher than men's.
- ◆ 🇨🇪 Switzerland and 🇨🇪 Czechia have **exceptional women's win rates**, despite smaller volumes.
- ◆ 🇪🇸 Some countries like Spain or 🇩🇰 Denmark show **lower win rates**, suggesting room for deeper exploration.

► Win Rate vs Total Games Played

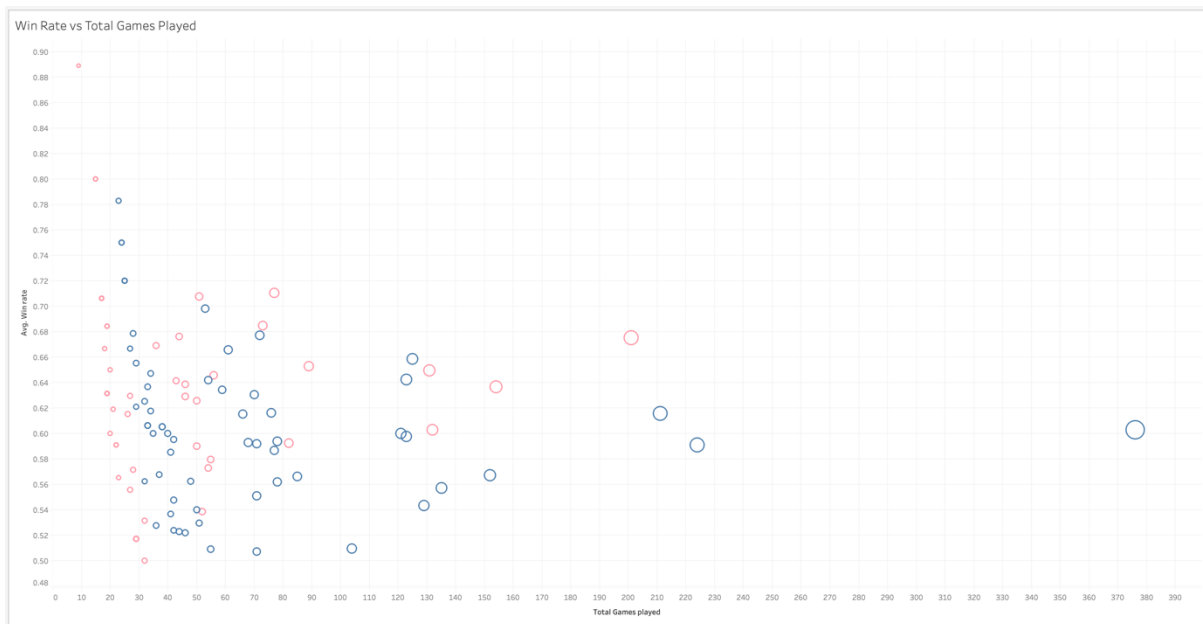


Figure 16: Win Rate vs Total Games Played (Performance Distribution)

- ◆ **X-axis:** Total Games Played
- ◆ **Y-axis:** Average Win Rate
- ◆ **Colour:** Gender (● Men, ● Women)
- ◆ **Size:** Number of Games Played (bigger dot = more matches)

🔍 Key Observations

1. Most players are concentrated between **20 to 150 games**, with win rates generally ranging from **0.55 to 0.70**.
2. A few players with **over 200 or even 350+ games** (large dots far right) show only **moderate win rates**, suggesting that **more matches don't guarantee higher win rates**.
3. Some players on the **upper-left area** of the chart have **fewer games but high win rates**, which may indicate **dominant short-term performance or peak efficiency**.
4. Female players (pink dots) appear more **tightly grouped** with slightly higher win rate clusters, possibly due to format differences (best-of-3 matches) or **more consistent performance across fewer games**.

💡 Insights Summary

- ◆ Players with **shorter careers and high win rates** stand out as highly efficient.
- ◆ Long careers often result in **averaged-out performance**, possibly due to aging, changing competition, or stamina.
- ◆ The gender distribution hints that **women might maintain stronger short-term performance**, while **men show wider variance**, possibly reflecting the 5-set format and higher match count.

4.7 Game Pattern Analysis (Set-by-Set Performance)

► Champions' Set-by-Set Win Rate Patterns

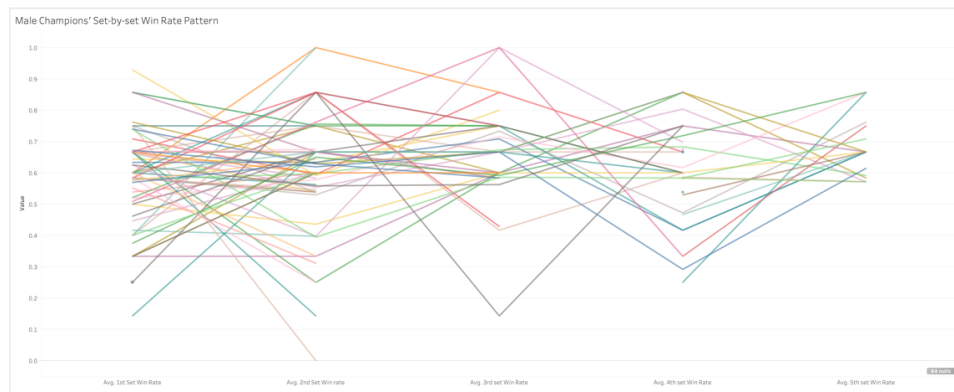


Figure 17: Male Champions' Set-by-Set Win Rate Patterns

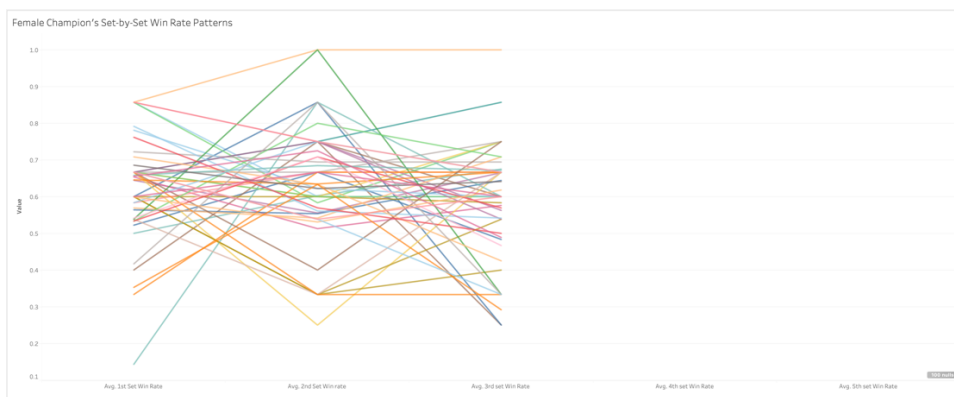


Figure 18: Female Champions' Set-by-Set Win Rate Patterns

🧑 Male Champions (Figure 17)

Win rates across five sets show **more fluctuations**, reflecting **greater volatility** and strategic adaptation. Some players recover from early set losses, while others fade in later sets due to fatigue. The five-set format makes men's matches more **unpredictable and demanding**.

🧑 Female Champions (Figure 18)

Female players show **more consistent performance** across three sets. A minor dip in the 2nd set may suggest mid-match adjustment, but overall, their win patterns are **steadier**, likely due to the shorter format minimizing endurance effects.

💡 Insight Summary

Men's matches are **more competitive and variable**, especially in longer matches.

Women's matches show **stronger early momentum** and **more uniform performance**.

5. Conclusion

1. Storytelling Structure & Visualisation Techniques

This report followed a **narrative-driven approach**, starting from global participation trends and narrowing down to individual performance analysis. A diverse range of visualisation types—**maps, scatter plots, treemaps, line charts, and Gantt charts**—were strategically selected and **consistently colour-coded by gender**. This ensured clarity, accessibility, and effective comparison across metrics. Tableau’s interactivity further enhanced the depth of analysis, enabling a **high level of visual storytelling**.

2. Gender-Based Trends & Key Differences

Men’s matches, following a best-of-five format, showed **higher volatility and more fluctuating win rate patterns**, particularly in the later sets. This reflects **greater physical and mental demands**, and less predictable outcomes.

In contrast, **women’s matches revealed more consistent and stable win rates** across three sets, likely due to the shorter format.

Geographic trends also highlighted **gender-based dominance**—e.g., **Australia and the U.S.** produced more female champions, while **Serbia and Switzerland** showed stronger male dominance.

3. Top Player Performance Analysis

Key players such as **Margaret Court, Novak Djokovic, and Serena Williams** stood out not only for their **title counts**, but also for maintaining **competitive win rates** over long careers. However, high win counts didn’t always align with win rate dominance, suggesting that **early momentum (in women’s matches)** and **late-set stamina (in men’s matches)** were crucial to outcomes.

Set-by-set line charts and win rate scatter plots revealed how **performance under fatigue or pressure** impacted match success.

4. Tableau’s Analytical Strengths

Tableau enabled multi-dimensional analysis of **gender, seed, country, win rate, and time**, providing insights at both macro and micro levels. The use of **interactive dashboards, filterable visuals, and layered data storytelling** made it possible to identify outliers, consistencies, and comparative strengths with ease.

5. Predictive Insight & Strategic Implications

Looking forward, several projections can be made

- ❑ **Countries with balanced gender success** (e.g., Australia, U.S.) may continue to be strong contenders for both male and female championships.
- ❑ **Moderately seeded players with high win rates** may represent emerging talents or underrated competitors.
- ❑ Given the **consistency in women’s win rates**, coaching strategies may emphasise **early-match momentum** and mental resilience.
- ❑ For men, **physical endurance and 4th/5th set stamina** remain critical performance differentiators, highlighting the value of conditioning.